

Analysis of Swarm Intelligence Metaheuristics for Mitigating Routing Attacks in VANETs

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Abstract—Vehicular Ad-Hoc Networks (VANETs) are characterized by high mobility, dynamic topologies, and strict Quality of Service (QoS) requirements, making them highly susceptible to routing attacks. This report aims to perform an analysis of three major Swarm Intelligence (SI) techniques: Ant Colony Optimization (ACO), Secretary Bird Optimization Algorithm (SBOA), and Hybrid Genetic Firefly Algorithm (HGFA) and their impact on routing security and network reliability. By evaluating related academic papers, this study compares their efficacy in attack detection and their impact on QoS and network performance metrics. The findings suggest that while ACO excels in ensuring route reliability and energy conservation, SBOA is best suited for system-wide security monitoring and cloud-integrated infrastructures, and HGFA is designed for maintaining link reliability in volatile urban settings.

I. INTRODUCTION

A. Background

VANETs are a subgroup of Mobile Ad-Hoc Networks (MANETs) designed to enable Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communication. The primary goal is to enhance road safety and traffic efficiency. However, the lack of a centralized infrastructure and the open nature of wireless communication invite malicious actors to disrupt the routing process. Traditional methods often struggle with the low-latency requirements of vehicular safety applications, leading researchers toward bio-inspired Swarm Intelligence (SI) to provide adaptive, decentralized security. SI is a branch of Artificial Intelligence (AI) inspired by the collective, decentralized behavior of social animals such as ants, bees, birds, and fish. In these systems, individual members (often called "agents") follow very simple, local rules. Even though there is no central "leader" or master plan, the group as a whole exhibits highly sophisticated, "intelligent" behavior that allows them to solve complex problems, such as finding the shortest path to food, escaping predators, or building intricate nests.

B. Scope of Review

This report focuses on the application of SI metaheuristics to the routing layer of VANETs. It specifically analyzes how

these algorithms identify malicious nodes and reroute traffic to maintain QoS. The scope is limited to a comparative analysis of three specific peer-reviewed papers representing ACO, SBOA, and HGFA methodologies.

II. PROBLEM DEFINITION

A. Problem Statement

The dynamic nature of VANETs creates a moving target for security protocols. Malicious nodes can easily join the network and execute routing attacks that drop, delay, or misdirect critical safety data. Existing standard routing protocols (like AODV or DSR) are trustworthy by nature and cannot inherently distinguish between a link failure due to high mobility and a deliberate attack.

B. Motivation

Each swarm algorithm has its advantages and disadvantages and can be applied to different networking scenarios. Motivation for this study stems from the need to categorize which SI technique is best suited for specific use cases.

III. BACKGROUND

A. VANETs

VANET stands for Vehicular Ad-Hoc Network. It is a specialized form of Mobile Ad-hoc Network (MANET) where the nodes are vehicles (cars, buses, trucks) as well as roadside infrastructure [5]. Instead of relying on a fixed, centralized hub such as a cell tower for every interaction, vehicles communicate with each other dynamically in motion [5]. A VANET consists of three main components:

- On-Board Units (OBU): Devices installed in vehicles which allow for the processing and transmission of data [5].
- Roadside Units (RSU): Devices located along the road acting as gateways [5].
- Communication Links: The Vehicle-to-Vehicle and Vehicle-to-Infrastructure communication mechanisms used to deliver messages in order to optimize traffic flow [5].

B. VANET Security Challenges

The security of VANETs presents a unique and high-stakes challenge because, unlike traditional data networks, a breach can lead to physical harm or loss of life. The primary difficulty lies in the network's extreme dynamism; vehicles move at high speeds, forming and breaking connections within seconds, which makes establishing a stable trust system incredibly complex. Security frameworks must simultaneously satisfy the CIA3 model (Confidentiality, Integrity, Authentication, Availability, and Access Control) while maintaining low latency such that if a safety message is delayed by even a few milliseconds, consequences could be extremely dire.

C. Advantages of Swarm Intelligence (SI) in VANETs

Swarm Intelligence (SI) leverages decentralized, bio-inspired behaviors to address the dynamic security and operational needs of VANETs. By utilizing collective intelligence, these algorithms provide robust, flexible solutions for evolving vehicular environments [5].

- Scalability: SI's decentralized nature prevents performance degradation as networks grow. This allows algorithms like Particle Swarm Optimization (PSO) to maintain high detection accuracy across varying vehicle densities, from rural roads to dense urban centers [5].
- Adaptability: SI thrives in high-mobility environments by enabling decisions in real-time. Ant Colony Optimization (ACO)-based protocols, for example, dynamically reroute data to bypass emerging malicious threats or sudden traffic congestion [5].
- Fault Tolerance: By relying on group decision-making rather than a central hub, SI ensures the network remains functional even if individual nodes fail. Bee-inspired algorithms can autonomously reroute traffic under adverse conditions without system-wide interruption [5].
- Decentralization: Eliminating central authorities removes single points of failure, reducing vulnerability to targeted attacks. This distributed approach manages trust and communication efficiently without creating processing bottlenecks [5].
- Resilience: Unlike static traditional frameworks, SI adapts to rapid topology changes, enhancing both security and resource allocation [5].

D. Routing Attack Types in VANET

To evaluate the effectiveness of SI, one must define the threats they are designed to mitigate:

- Black Hole Attack: A malicious node advertises the shortest path to a destination but drops all intercepted packets instead of forwarding them, leading to a massive drop in PDR [4].
- Wormhole Attack: Two colluding nodes capture packets at one point and tunnel them to another location, confusing the network topology [4].

- Grey Hole Attack: A variation of the Black Hole where the node selectively drops packets (e.g., only UDP packets or specific IDs), making detection difficult [4].
- Sinkhole Attack: An intruder tries to attract all traffic from a specific area through itself by spoofing routing information [4].
- Sybil Attack: A single node assumes multiple identities (pseudonyms) to simulate a traffic jam or outvote honest nodes in trust-based systems [4].
- Illusion Attack: The attacker broadcasts false sensor data (e.g., a fake accident) to manipulate surrounding vehicles [4].
- Denial of Service (DoS): Flooding the network with useless traffic to exhaust bandwidth and prevent safety messages from being delivered [4].
- Replay Attack: A malicious node deliberately stores and replays held obsolete information which negatively impacts traffic [4].
- Timing Attack: A malicious node deliberately delays the delivery of a packet carrying critical information, causing traffic accidents and jams [4].

IV. SELECTED ARTICLES

A. A Multi-Metric Stability Function based Intelligent Routing Protocol to Mitigate Network Layer Attacks in VANETs

Ant Colony Optimization (ACO) is inspired by the foraging behavior of ants. When ants find food, they leave a chemical trail called pheromones [5]. Other ants detect these trails and follow them, reinforcing the pheromone levels. In VANETs, ants are control packets searching for paths and pheromones represent path quality (low delay, high trust). If a path starts dropping packets (Black Hole), the pheromones evaporate, and the swarm naturally shifts to safer, more reliable routes [5].

- Summary: This paper addresses the lack of centralized control in VANETs, which leaves high-speed vehicular communication vulnerable to various disruptions. The authors propose Stability Function-based Intelligent AODV (Ad hoc On-Demand Distance Vector) Routing Protocol (SF-AODV), a hybrid framework incorporating SF with ACO. The framework is twofold: it first filters out malicious nodes (Flooding and Black Hole in particular) using real-time stability metrics, then it employs an updated ACO mechanism to select the most efficient and reliable routing path among remaining trusted nodes. Simulation results using NS-2 indicate this methodology outperforms traditional AODV based on routing metrics such as Packet Delivery Ratio (PDR), delay, and energy efficiency [1].
- Problem Formulation: The high mobility and decentralized nature of VANETs render them highly susceptible to network layer threats such as: 1) Flooding attacks - where intruders transmit an excessive number of control

packets to non-existent destinations, causing congestion and forcing packets to be dropped by other nodes (a form of DoS attack), 2) Black Hole attacks [1]. Frequent topology changes also lead to link failures and reduced network lifetime.

- **Methodology and Algorithm:** The methodology follows a two-stage filter then optimize protocol layered on top of the traditional AODV framework.

- 1) **Stage I: Node Filtering via SF:** The algorithm evaluates each node i using a composite stability score:

$$SF_i = w_1 \cdot S_{TC_i} + w_2 \cdot S_{TD_i}$$

Where S_{TC} is the Traffic Congestion quantified by the ratio of current load to the node's maximum capacity and S_{TD} is the Delay Time quantified as the time taken for a vehicle to receive and send a routing packet. A node is flagged if its congestion exceeds a threshold (C_{th}) of 70% of its capacity or if its delay time deviates significantly from expected radio range values [1].

- 2) **Stage II: ACO Route Optimization:** After filtering unstable nodes, the source broadcasts a modified RREQ containing fields for traffic congestion, stability, and battery power. The destination then calculates a Pheromone Value (P) to select the optimal path:

$$P = \frac{(SF_i)(RBP_i)}{(HC_i)(RTT_i)}$$

Where RBP is the Residual Battery Power, HC is the Hop Count, and RTT is the Round Trip Time. The path with the highest cumulative pheromone is selected [1].

- **Simulation Environment:** The authors utilized Network Simulator (NS)-2.34 to validate the protocol.

- 1) **Parameters:** A simulation area of 2500m x 2500m with 100 vehicles using the 802.11p MAC protocol [1].
- 2) **Scenarios:** The performance was tested by modifying vehicle density (20 to 100 nodes) and vehicle speed (5m/s to 25 m/s) [1].
- 3) **Comparison:** SF-AODV was benchmarked against SVODR, SRMAD, IM-AODV, and traditional AODV [1].

- **Key Results:**

- 1) **Traffic Congestion:** SF-AODV showed approximately 51% less congestion than benchmarked methods at high densities [1].
- 2) **Delay Reduction:** The delay metric in SF-AODV was roughly 77.7% lower than benchmarked methods and 29.4% lower than the nearest competitor (SVODR) [1].
- 3) **Energy Efficiency:** SF-AODV maintained significantly higher residual battery power across all

speeds and densities by reducing the transmission of unnecessary control packets [1].

- 4) **Attack Mitigation:** The stability function effectively localized and bypassed nodes involved in Flooding and Black Hole behaviors [1].

- **Conclusion:** The paper concludes that the multi-metric SF-AODV protocol successfully mitigates network layer attacks while adapting to the dynamic nature of VANETs. By prioritizing nodes with high stability and sufficient energy, the protocol ensures superior QoS and network longevity.

- **Analysis and Critique:**

- 1) **Strengths:**

- a) Integrating battery power (RBP) into the ACO pheromone calculation is a vital addition for VANETs, as routing efficiency is meaningless if the chosen nodes are near exhaustion.
- b) The two-stage approach (filtering via SF then optimizing via ACO) provides a proactive defense that prevents malicious nodes from even entering the path-finding stage.

- 2) **Critique/Limitations:**

- a) The protocol assigns equal weights ($w_1, w_2 = 0.5$) to congestion and delay. In high-speed urban environments, delay might be a more critical indicator of a Black Hole attack than simple congestion, suggesting that dynamic weighting might be more effective.
- b) Calculating SF and P values for every neighbor in a 100-node high-speed swarm may introduce processing latency that is not fully explored in the simulation discussion.
- c) It is unclear how the SF framework would treat Gray Hole attacks and differentiate them from Black Hole attacks.

B. Advanced DDoS Attack Detection in VANET Cloud Systems Using Secretary Bird Optimization and Machine Learning

The Secretary Bird Optimization Algorithm (SBOA) is a novel bio-inspired metaheuristic based on the predatory behaviors and survival strategies of secretary birds. Like other swarm intelligence methods (e.g., ACO or PSO), SBOA utilizes a population of search agents to navigate complex multidimensional spaces. It is characterized by two distinct phases: Exploration (Survey Phase) and Exploitation (Strike Phase) [2]. During the exploration phase, agents survey the search space broadly to maintain population diversity and prevent the algorithm from getting stuck in local optima. During the strike phase, agents perform rapid, focused movements toward promising regions to refine the search and converge on a global optimum [2]. In this paper, SBOA is used as a hyperparameter tuner to find the most efficient configuration for Machine Learning (ML) classifiers.

- **Summary:** The paper proposes an optimized Intrusion Detection System (IDS) that combines SBOA with a

Random Forest (RF) classifier to detect Distributed Denial of Service (DDoS) attacks in real-time. The research addresses the limitations of traditional ML models, which often suffer from high false-alarm rates and difficult manual tuning. By integrating SBOA, the system automatically balances multiple objectives: accuracy, detection time, PDR, and energy consumption. The resulting SBOA-RF model achieves near-perfect detection rates, significantly outperforming standard models and existing swarm-based approaches like PSO and Genetic Algorithm (GA) [2].

- **Problem Formulation:** The study identifies DDoS attacks as a critical risk to Intelligent Transportation Systems (ITS). These attacks flood the network with malicious traffic, which: 1) blocks essential safety messages and emergency alerts, 2) causes measurable damage to network performance, including degraded PDR, and 3) leads to increased network latency and accelerated energy depletion of vehicular nodes. The research specifically targets the problem of hyperparameter optimization in ML classifiers, which is typically time-consuming and prone to error when performed manually [2].

- **Methodology and Algorithm:** The methodology employs a six-layer system architecture for detection and response:

- 1) **Traffic Monitoring and Feature Extraction:** Raw data is collected via NS2 and processed into features such as PDR, energy consumption, IP addresses, and timestamps [2].
- 2) **SBOA Optimization Engine:** 30 agents iterate 100 times to optimize the RF parameters (e.g., number of estimators and max depth) [2].
- 3) **Multi-Objective Fitness Function:** SBOA evaluates agents based on a weighted formula [2]:

$$F(X) = w_1 \cdot \text{Accuracy} + \frac{w_2}{\text{Time}} + w_3 \cdot \text{PDR} + \frac{w_4}{\text{Energy}}$$

- 4) **Classification and Response:** The optimized model classifies traffic. If an attack is detected with a confidence score > 0.85 , the system blocks the attacker, alerts the RSU/Cloud, and reroutes traffic [2].

- **Simulation Environment:** The authors used Network Simulator 2 (NS2) for the experimental setup:

- 1) **Parameters:** A simulation area of 2000m x 500m highway/urban road [2]. 20–50 vehicles traveling at speeds of 60–100 km/h [2].
- 2) **Scenarios:** 10–25% of the network nodes were malicious. Attack types included UDP flood, TCP SYN flood, and application-layer HTTP flood. There were 200,000 samples with a 50/50 split between normal and attack traffic [2].

- **Key Results:**

- 1) **Detection Accuracy:** SBOA-RF achieved 99.78% accuracy, surpassing baseline RF (99.47%) and

Decision Tree (99.12%) models [2].

- 2) **Competitive Benchmark:** The SBOA-based method outperformed existing PSO-based (99.65%) and GA-based (99.58%) approaches [2].
- 3) **Reliability:** The model showed a balanced error distribution with only 88 misclassifications out of 40,000 test samples [2].
- 4) **Real-Time Speed:** The system maintains an average response time of 0.73 seconds and an inference time of 0.012 seconds per sample [2].
- 5) **Error Reduction:** Optimization reduced false positives and false negatives by over 58% compared to standard RF [2].

- **Conclusion:** The research concludes that the SBOA-RF framework is a highly effective and practical solution for protecting VANET-Cloud systems. It provides a robust defense against various DDoS strategies while ensuring minimal disruption to legitimate communication. The low memory footprint (156 MB) and rapid inference time make it suitable for deployment on resource-constrained RSUs and edge devices.

- **Analysis and Critique:**

- 1) **Strengths:**

- a) Unlike many papers that focus solely on accuracy, this study incorporates energy consumption and latency into the swarm's fitness function, which is critical for real-world automotive hardware.
- b) Testing against HTTP floods (which mimic normal traffic patterns) demonstrates the model's sophistication in handling application-layer threats that often bypass simpler network-layer filters.

- 2) **Critique/Limitations:**

- a) While 0.73 seconds is fast for a cloud response, in a high-speed collision-avoidance scenario (100 km/h), a vehicle travels nearly 20 meters in that time. For safety-critical beacons, even lower latency may be required.
- b) The simulation uses AODV on a 2000m x 500m area. While this is standard, the performance of the SBOA-RF model in a highly dense, 3D urban environment with buildings (obstructions) and more complex mobility traces would be a valuable next step for validation.

C. Hybrid Genetic Firefly Algorithm-Based Routing Protocol for VANETs

The Firefly Algorithm mimics the flashing behavior of fireflies to attract mates or prey [6]. In VANETs, brightness represents a node's trust level or signal strength. Dimmer fireflies move toward brighter ones. Malicious nodes (Black Hole/Sybil) are assigned a low brightness because of their poor performance. Legitimate nodes are naturally attracted to

bright, high-performing nodes, creating secure clusters that isolate intruders [6].

- **Summary:** The paper proposes the Hybrid Genetic Firefly Algorithm (HGFA) to overcome common limitations in standard swarm intelligence, such as slow convergence and getting stuck in local optima [3]. By integrating Genetic Algorithm (GA) features like crossover and mutation into the firefly framework, the protocol adaptively optimizes routes in both sparse and dense vehicular environments [3]. The approach is designed to maintain high performance in various traffic scenarios, specifically targeting the need for reliable communication in high-mobility networks.
- **Problem Formulation:** The research identifies that while VANETs are critical for road safety and traffic management, they suffer from several inherent challenges:
 - 1) **Intermittent Connectivity:** High vehicle speeds lead to frequent link failures and topology changes [3].
 - 2) **Algorithmic Limitations:** Standard metaheuristics like PSO or basic FA often suffer from premature convergence, leading to suboptimal routing solutions [3].
 - 3) **Varying Densities:** Protocols often struggle to maintain efficiency across both high-density urban areas and low-density highways [3].
 - 4) **Security Concerns:** The paper highlights that routing security and node management are critical hurdles for deploying V2V communication systems [3].
- **Methodology and Algorithm:** The HGFA integrates GA features into the firefly process through a dual-layered optimization approach:
 - 1) **GA Initialization:** A Genetic Algorithm initializes the population and uses crossover to generate diverse candidate paths [3].
 - 2) **Fitness Evaluation:** Paths are evaluated based on a multi-metric function that considers retransmission occurrence (N_R) and propagation time (T_P) [3].
 - 3) **Firefly Movement:** Vehicles (fireflies) update their positions toward brighter (more fit) neighbors based on Cartesian distance and light intensity [3].
 - 4) **Adaptive Mutation:** If the swarm stops finding better paths, GA mutation is triggered to jump the search to a different part of the network topology, avoiding local optima [3].
- **Simulation Environment:** The authors validated the HGFA using Network Simulator 3 (NS-3.26):
 - 1) **Parameters:** Tested in sparse networks (8–50 nodes) and dense networks (up to 500 nodes). Vehicle speeds varied between 10 and 60 km/h [3].
 - 2) **Scenarios:** Utilized the Manhattan Mobility Model to simulate realistic city grid traffic patterns [3].
 - 3) **Comparisons:** Compared HGFA against standard Firefly, PSO, AODV, and OLSR protocols [3].

- **Key Results:**

- 1) **Performance Gain:** HGFA outperformed PSO by 0.55% in dense scenarios and 0.42% in sparse scenarios regarding transmission time [3].
- 2) **Efficiency:** The protocol showed significantly lower transmission times and lower routing overhead compared to traditional AODV and OLSR [3].
- 3) **PDR:** In sparse network tests, HGFA achieved a PDR of 41%, compared to 28% for AODV and 34% for GA-ACO [3].
- 4) **Reliability:** The hybrid components effectively prevented the local optima trap, allowing for faster and more consistent path discovery across all tested densities [3].

- **Conclusion:** The authors conclude that the HGFA protocol is a robust and future-proof alternative for VANET routing. By balancing the global exploration capabilities of GA with the local exploitation of the Firefly Algorithm, the system ensures reliable and optimized communication. The protocol is proven to be effective across various traffic conditions, meeting the requirements for next-generation intelligent transportation systems.

- **Analysis and Critique:**

- 1) **Strengths:**

- a) The mathematical integration of GA's mutation features directly addresses the stagnation or premature convergence issues found in standard swarm-based papers.
- b) Testing across a wide range of node densities (8 to 500) provides strong evidence for the protocol's scalability in real-world urban environments.

- 2) **Critique/Limitations:**

- a) While the paper mentions security and node management as motivations, it focuses primarily on QoS metrics (PDR, delay) rather than testing against specific adversarial models like Black Hole or Sybil attacks found in other security-focused literature. A Black Hole node could mimic high brightness to attract traffic as well - the paper could have integrated a trust score into the brightness metric to prevent fireflies from being lured by malicious nodes.
- b) The simulation's maximum speed of 60 km/h is appropriate for urban settings but does not verify the protocol's stability at high highway speeds where link decay is much more rapid.
- c) The paper does not provide an in-depth analysis of the additional processing overhead introduced by running both GA and FA on a vehicle's onboard unit.

V. CONCLUSION

A. Summary

This research report conducted a comparative analysis of three advanced swarm intelligence (SI) metaheuristics—Ant Colony Optimization (ACO), Secretary Bird Optimization Algorithm (SBOA), and the Hybrid Genetic Firefly Algorithm (HGFA)—to evaluate their effectiveness in mitigating routing-layer and system-level attacks in Vehicular Ad-Hoc Networks (VANETs). The SF-AODV protocol utilizes a multi-metric stability function to filter nodes before employing ACO for path selection, successfully reducing network congestion by approximately 51% while mitigating Black Hole and Flooding attacks. The SBOA-RF framework integrates bio-inspired hyperparameter tuning with machine learning to achieve a 99.78% detection accuracy against Distributed Denial of Service (DDoS) threats, maintaining sub-second response times critical for vehicular safety. Finally, the HGFA protocol addresses the premature convergence trap of standard SI methods by hybridizing Firefly movement with Genetic Algorithm mutation, nearly doubling the PDR of traditional protocols in sparse network environments.

B. Discussion

In the context of routing security and network reliability, the three methods explored in this research report can be applied to different use cases.

- 1) Ant Colony Optimization (ACO) is most effectively deployed as a proactive node-filtering mechanism at the network layer. The best use case is high-speed scenarios requiring internal node integrity and energy awareness.
- 2) Secretary Bird Optimization Algorithm (SBOA) is best utilized as a high-precision optimizer for system-wide Intrusion Detection Systems (IDS). The best use case is real-time volumetric attack detection in cloud-integrated or edge-computing VANET infrastructures.
- 3) Hybrid Genetic Firefly Algorithm (HGFA) is optimized for path recovery and link stability in the most volatile vehicular environments. The best use case is maintaining connectivity in complex urban grids with increased topological changes.

Something to note is that although both ACO and FA may seem similar in that they use attraction to find optimal paths, they do differ in their mechanism of communicating information and handling topology changes.

C. Conclusive Remarks

The findings demonstrate that while traditional routing protocols like AODV are highly efficient, their lack of inherent security makes them easy targets for malicious actors in decentralized environments. Swarm intelligence offers a dynamic solution by mimicking collective biological resilience to localize and bypass attackers. A key observation across these studies is that hybridization (e.g., GA with Firefly or SBOA with Random Forest) is superior to

standalone metaheuristics, as it balances global exploration with local exploitation, preventing the system from settling on suboptimal or compromised routes. Furthermore, the transition toward multi-objective fitness functions—which consider not just path length but also residual battery power, trust scores, and computational latency—is essential for the practical deployment of next-generation Intelligent Transportation Systems (ITS).

D. Future Work

While the evaluated metaheuristics show significant promise, several avenues remain for future exploration:

- Real-World Prototyping: Current results are largely based on simulation environments like NS-2 and NS-3; testing these algorithms on physical OBUs is necessary to validate their performance under real-world signal obstructions and hardware processing constraints.
- Complexity and Scalability: Future research should investigate the computational tax of running hybrid optimization models, ensuring that the time taken to optimize a route does not exceed the strict millisecond-latency requirements of life-safety safety beacons.
- Collaborative Security: Integrating these SI methods with Blockchain-based trust management or Federated Learning could allow vehicles to share threat intelligence anonymously, enhancing collective immunity against coordinated swarm-on-swarm attacks.
- Adversarial AI: As security protocols become more intelligent, attackers may employ adversarial ML to trick SI sensors; developing robust metaheuristics capable of identifying manipulated pheromones or brightness levels is a critical next step for 2026 and beyond.

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